SOUTH EASTERN UNIVERSITY OF SRI LANKA

SWT22022 Practical for Internet Application Development

Proposal for CA1 Group Assignment

Project Title	FoodBridge: A Smart Food Rescue and Waste Management Platform
Group ID	Group I
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1. Project Overview

FoodBridge is a proposed full-stack web platform that connects surplus-food donors (wedding halls, hotels, restaurants, bakeries, event organisers and individuals) with verified food recipients (NGOs, charities and shelters), volunteers who carry out pickup and delivery, and registered waste collectors and recycling/composting facilities who handle anything that is not fit for human consumption.

Instead of treating food rescue and waste disposal as two separate problems, the platform classifies every surplus post into either a food donation or a specific waste category at the point it is created, and then routes it to the correct destination through district-based and smart matching. The group intends to build the system as a MERN-stack (MongoDB, Express.js, React.js, Node.js) web application with five role-based dashboards: Donor, Recipient, Volunteer, Recycler/Waste Collector and Admin.

2. Problem Statement

Large-scale food preparation at weddings, hotels, restaurants and public events regularly produces both edible surplus food and recyclable or organic waste. At present, there is no simple, centralised way to route either of these to the right place in time, which creates the following recurring problems:

- Food donors often do not know which verified NGOs or shelters are nearby and available to receive surplus food before it spoils.
- Recipient organisations have no real-time visibility of surplus food in their area and rely on informal phone calls or personal contacts.
- There is usually no organised volunteer network to physically move food from a donor to a recipient within a safe time window.
- When food cannot be donated, donors do not know which licensed recycler or composting facility to contact, so recyclable materials (plastic, glass, paper, metal) and organic waste are often sent to general waste instead of being processed responsibly.
- Manually separating waste into the correct categories is time-consuming for donors and inconsistent without guidance.

This is both a social problem — edible food going to waste while people in need go without it — and an environmental problem, since recyclable and organic waste that could be diverted ends up in landfills. It affects food-service businesses, event organisers, community organisations, and, indirectly, the wider community through the associated social and environmental cost. Low-resource NGOs and small recycling operators are particularly affected, since they typically lack the logistics network to discover and collect what they need.

3. Proposed Solution

FoodBridge gives donors a single "surplus post" workflow that automatically branches to the right pathway. When a donor logs a surplus item, the system captures a small set of structured details (type, quantity, district, condition, pickup deadline) and uses this to classify the post as donatable food or as a specific waste category.

Verified recipients see nearby donatable food ranked by district, distance and freshness window. Verified recyclers and composting facilities see nearby waste-pickup requests filtered by material type. Volunteers see open pickup jobs they can accept and are guided by a map and route view. District-based and smart matching reduce the manual effort donors and recipients currently spend searching for each other, automatic notifications keep every party updated as a request moves through its lifecycle, and an admin verification layer keeps only genuine NGOs and recyclers active on the platform.

An impact dashboard turns individual transactions into cumulative, reportable figures — meals rescued, kilograms recycled, kilograms of organic waste diverted — which donor organisations and NGOs can reuse for their own reporting. The platform does not itself guarantee food safety; final acceptance of any food donation remains the responsibility of the receiving organisation. What the platform removes is the discovery, matching and logistics friction that currently prevents surplus from reaching the right destination.

4. Project Objectives

Main Objective

To design and develop a full-stack web platform that reduces food waste and improves recyclable and organic waste management by connecting donors, verified recipients, volunteers and waste collectors through district-based, smart matching.

Specific Objectives

- To allow donors to register surplus food or waste with structured details (type, quantity, district, photo, pickup deadline) through a short, guided posting flow.
- To automatically classify each posted item into a food donation or a specific waste category (plastic, glass, paper/cardboard, metal, organic, non-recyclable).
- To match each post to the nearest verified recipient or collector using district, distance, type, quantity and pickup deadline, and to surface the closest available matches.
- To provide a volunteer workflow that lets a pickup job be accepted, tracked and marked delivered/collected, with every status change time-stamped.
- To provide an admin module that verifies organisations, monitors platform activity, manages complaints/fake accounts and produces reports.
- To provide a public impact dashboard that aggregates meals rescued, waste recycled and organic waste diverted from real platform activity.

5. Target Users / Stakeholders

The system has the following main users and stakeholders:

User / Stakeholder	Role / Description	Main Needs
Food Donor	Wedding halls, hotels, restaurants, bakeries, event organisers and individuals with surplus food or waste.	A quick way to post surplus with minimal effort, visibility of who is collecting it, and confidence it will be collected before it spoils or before the deadline.
Food Recipient	Verified NGOs, charities, community organisations and approved shelters.	Real-time visibility of nearby available food, a simple request/accept flow, and trust that donor and food details are genuine.
Volunteer	Individuals who pick up and transport food or recyclable waste.	A clear list of nearby pickup jobs, map/navigation support, and confirmation of what to deliver, to whom, and by when.
Waste Collector / Recycler	Registered recycling businesses and composting facilities (plastic, glass, paper, metal, organic).	Waste requests filtered by the material types they handle, and the ability to accept jobs within their capacity and service area.

User / Stakeholder	Role / Description	Main Needs
Platform Administrator	Staff responsible for platform integrity.	Tools to verify organisations, monitor donation/waste activity, manage complaints and fake accounts, and generate reports.

6. Proposed Technology Stack

The group proposes a MERN-stack implementation, as required for a full-stack web application:

Layer / Technology	Proposed Choice & Brief Justification
Frontend	React.js, using React Router for navigation, component-based dashboards for each of the five roles, form validation, and a map component for pickup/location display. Chosen for its component reuse across five distinct dashboards.
Backend / API	Node.js + Express.js REST API handling authentication, donation/waste CRUD, the matching engine, pickup tracking, notifications and admin operations. Pairs naturally with React and MongoDB (MERN) and is covered in this course.
Database	MongoDB with Mongoose ODM. A document database fits the varying shapes of Donations, WasteRequests and Organization records without a rigid relational schema, and scales well across many independent collections.
Authentication / Security	JWT-based authentication with role-based access control (Donor, Recipient, Volunteer, Recycler, Admin), password hashing with bcrypt, and route-level authorisation middleware.
Other Tools / Libraries	A maps/geolocation API (e.g., Google Maps or Leaflet with OpenStreetMap) for district matching and route display; a cloud image-storage service for donation photos; an email/SMS or in-app notification service; Git/GitHub for version control.

7. Major Functional Modules

The project contains five substantial functional modules, each owned primarily by one group member (additional modules can be added if the group has more than five members):

Module	Description / Main Functions	Primary Member	Expected Complexity / Contribution
Module 1 — Donor & Surplus Posting	Donor registration; create food or waste postings with structured details (type, quantity, district/location, available time, pickup deadline, preparation time for food, photo, special instructions); edit/cancel active postings; posting history and status.	Member 1 (Group Leader)	High — multi-step form, file upload, validation, status lifecycle.
Module 2 — District-Based & Smart Matching (Recipient)	District → city/area → nearby-organisation search; ranks nearby verified recipients/collectors by district, distance, type, quantity and deadline; recipient dashboard to browse, request and accept nearby donations.	Member 2	High — core matching algorithm plus a full recipient-facing dashboard.

Module	Description / Main Functions	Primary Member	Expected Complexity / Contribution
Module 3 — Smart Waste Classification & Recycler/Collector	Classifies posted waste into plastic / glass / paper-cardboard / metal / organic / non-recyclable pathways and routes each to the appropriate recycler, composting facility or waste-management pathway; collector dashboard to view, filter and accept pickup requests by material type.	Member 3	High — classification logic plus a second full role-based dashboard.
Module 4 — Volunteer, Pickup & Route Tracking	Volunteers view and accept open pickup jobs; a shared status pipeline (Posted → Matched → Pickup Scheduled → Picked Up → Delivered/Recycled → Completed) with timestamps; map view of donor/recipient/recycler/volunteer locations and the pickup route.	Member 4	Moderate–High — status-machine logic, map/route integration, live tracking.
Module 5 — Admin, Verification, Notifications & Impact Dashboard	Admin verification of NGOs/recyclers, monitoring of all donations and waste requests, complaint handling and fake-account removal, report generation; automatic notifications for every role at key status changes; a public impact dashboard aggregating meals rescued, kg recycled, kg organic waste diverted and districts covered.	Member 5	High — combines admin CRUD/verification, a notification layer touching every module, and reporting/aggregation.

8. Individual Contribution Plan

Each member is mapped to one major module, with concrete tasks and deliverables that can be assessed individually:

Member / Reg. No.	Assigned Module / Responsibility	Key Tasks	Expected Deliverables
Member 1 (Group Leader) [Reg. No.]	Module 1 — Donor & Surplus Posting	Design and build donor registration and posting forms; implement posting CRUD API and status lifecycle; coordinate integration across modules.	Working Donor module (UI + API) with test cases; integrated build coordination notes.
Member 2 [Reg. No.]	Module 2 — District-Based & Smart Matching (Recipient)	Design the district/distance-based matching algorithm; build the recipient dashboard and request/accept flow.	Working matching engine and Recipient module (UI + API) with test cases.
Member 3 [Reg. No.]	Module 3 — Smart Waste Classification & Recycler/Collector	Design the waste-classification logic and category-to-pathway routing; build the recycler/collector dashboard.	Working waste classification logic and Recycler module (UI + API) with test cases.

Member / Reg. No.	Assigned Module / Responsibility	Key Tasks	Expected Deliverables
Member 4 [Reg. No.]	Module 4 — Volunteer, Pickup & Route Tracking	Build the volunteer dashboard, pickup status pipeline, and map/route view.	Working Volunteer module (UI + API) with test cases; map integration demo.
Member 5 [Reg. No.]	Module 5 — Admin, Verification, Notifications & Impact Dashboard	Build the admin verification/reporting tools, the notification layer, and the impact dashboard aggregation.	Working Admin module (UI + API) with test cases; impact dashboard demo.

9. Key Functional Requirements

The system must support the following key functions:

- Role-based registration and login for Donor, Recipient, Volunteer, Recycler/Waste Collector and Admin accounts.
- Creation of a food donation or waste post with type, quantity, district/location, available time, pickup deadline, preparation time (for food), photo, and special instructions.
- Automatic classification of each post into a donation or a specific waste category (plastic, glass, paper/cardboard, metal, organic, non-recyclable).
- District-based and smart matching that ranks nearby verified recipients or collectors by district, distance, type, quantity and deadline.
- Request/accept workflow for recipients and collectors, and a job-accept workflow for volunteers.
- A shared pickup-status pipeline with timestamps, visible to the donor, recipient/recycler and volunteer involved in that request.
- Map display of donor, recipient, recycler and volunteer locations, with the pickup route shown for an active job.
- Automatic notifications for key events (donation accepted, new nearby donation, new pickup request, pickup assigned).
- Admin verification of NGOs and recyclers, with a visible "Verified" badge for approved organisations.
- Admin tools to monitor donations/waste requests, manage complaints, and remove fake accounts.
- Report generation and a public impact dashboard (meals rescued, kilograms recycled, organic waste diverted, districts covered).
- Full history/tracking view for each donor, recipient and collector of their past requests.

10. High-Level System Architecture

The system follows a standard three-tier MERN architecture. The React.js client renders a different dashboard per role but talks to a single Express.js REST API over JSON. The API layer contains the business logic — authentication and role-based access control, the donation/waste service, the matching engine, pickup/tracking, notifications and admin/reporting — and is the only layer that talks directly to MongoDB via Mongoose. External services (a maps/geolocation API, cloud image storage, and an email/SMS notification

gateway) are called from the backend rather than directly from the client, so that API keys and business rules stay server-side.

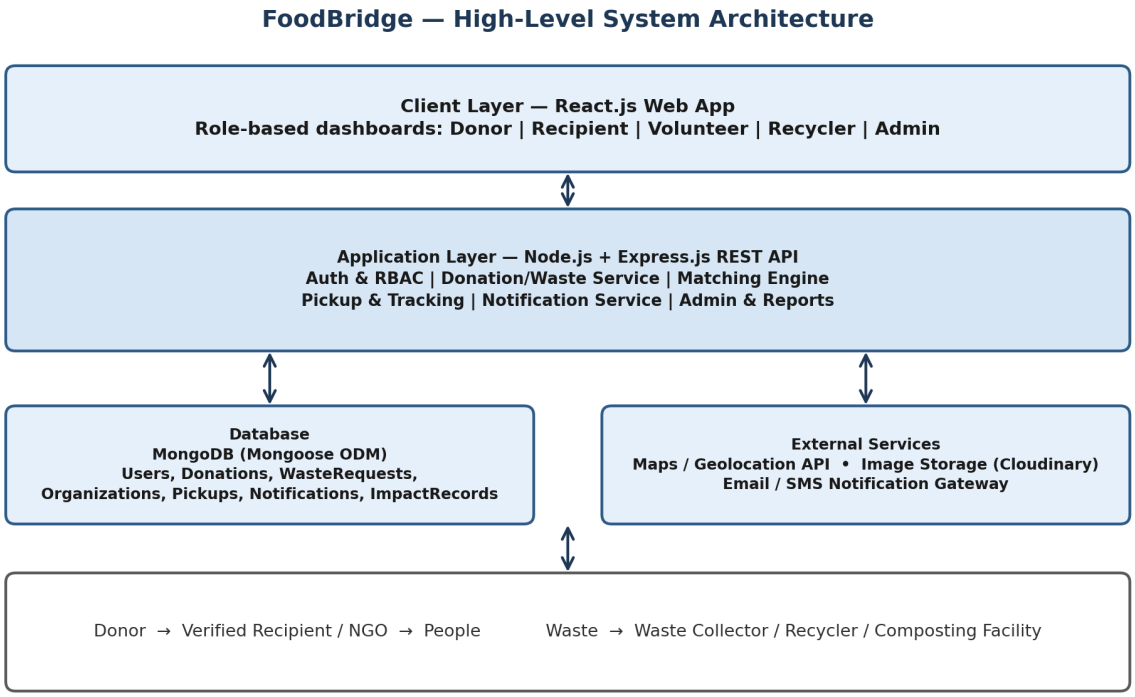


Figure 1: FoodBridge high-level system architecture

This separation keeps the frontend simple (it only renders data and calls the API) while all matching, classification and verification logic lives in one place on the backend, which will make it easier for five members to work on different modules of the same API without conflicting with each other's frontend code.

11. Preliminary Database Design

At proposal stage, the group anticipates the following main MongoDB collections:

Entity / Collection	Purpose	Important Data
Users	Stores account and profile data for every role.	name, email, password hash, role (donor/recipient/volunteer/recycler/admin), district, contact number, verification status.
Organizations	Profile and verification record for NGOs, charities and recycling/composting facilities.	organisation name, type, registration/license info, district, verification status, linked user account.
Donations	One record per food-surplus post.	donor ID, food type, quantity, prepared time, storage condition, pickup deadline, photo, district, status.
WasteRequests	One record per waste-collection post.	donor ID, waste category (plastic/glass/paper/metal/organic/non-recyclable), quantity, district, pickup deadline, status.

Entity / Collection	Purpose	Important Data
Pickups	Tracks the pickup/delivery job linking a donor, a volunteer, and a recipient or recycler.	donation/waste request ID, volunteer ID, recipient/recycler ID, status pipeline stage, timestamps, route/location.
Notifications	Stores in-app/email/SMS notifications sent to users.	recipient user ID, message, related donation/waste/pickup ID, read status, timestamp.
ImpactRecords	Aggregated figures for the public impact dashboard.	meals rescued, kg recycled (by category), kg organic waste diverted, districts covered, period.

12. Proposed Development Plan

The group proposes the following 14-week plan, aligned to the course schedule. Module development for Modules 1–3 overlaps slightly with Modules 4–5 once the core API and matching logic are stable, so the five members can work in parallel after week 4.

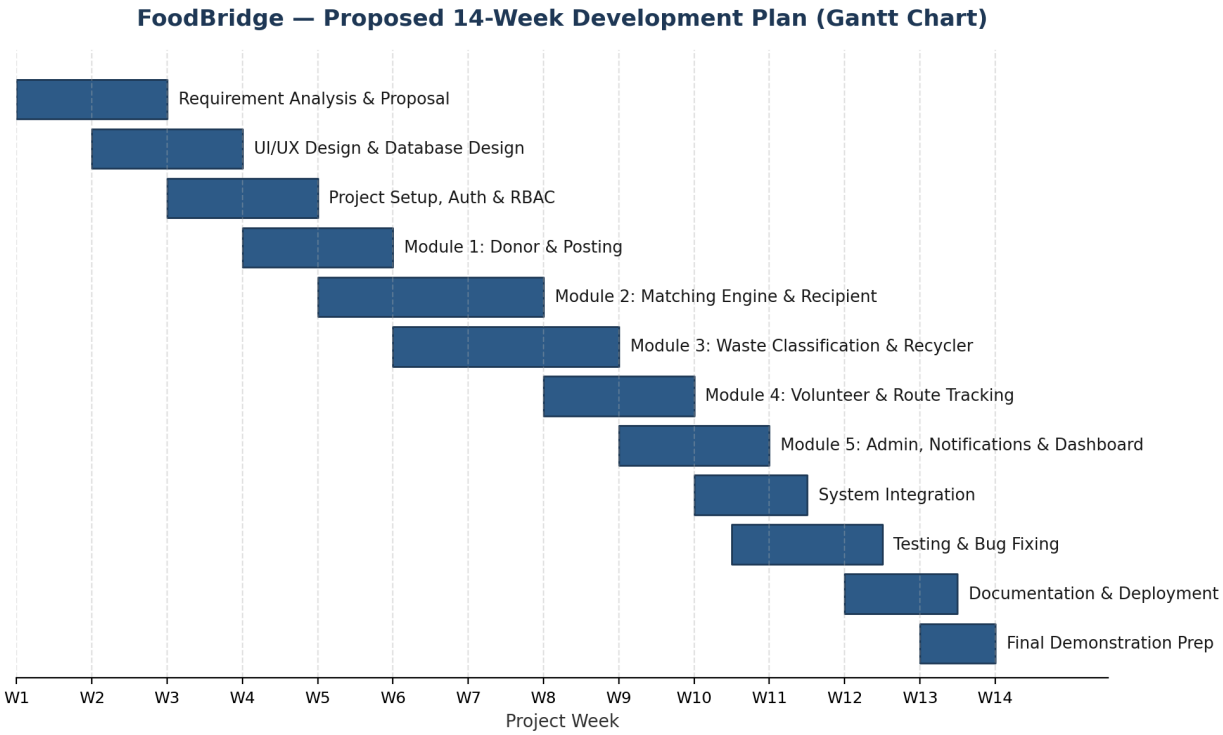


Figure 2: Proposed 14-week development plan (Gantt chart)

13. Expected Outcome

The completed system is expected to be a working full-stack MERN platform that demonstrates district-based and smart matching, correct classification of waste into the right pathway, five role-based dashboards, a tracked pickup lifecycle, and a measurable impact dashboard. Donors benefit from an easy, compliant way to donate or dispose of surplus; NGOs benefit from real-time visibility of nearby food; volunteers benefit from an organised task list; recyclers benefit from a steady stream of correctly-sorted material; and the wider community benefits from less food waste, more recycling, and a reusable record of the platform's social and environmental impact.

14. References / Supporting Sources

- Food and Agriculture Organization of the United Nations (FAO) — publications on global food loss and food waste, used to understand the scale and causes of the problem.
- Existing surplus-food-sharing applications such as Too Good To Go and OLIO, reviewed for inspiration on donor/recipient matching flows.
- Official documentation for React, Node.js/Express, and MongoDB/Mongoose, for the proposed technology stack.
- Local municipal/solid-waste-management guidance on waste categories, to be confirmed with local authorities during requirement analysis.